



G-PEARL v1.0 developed by Dr. Simone Zocca and Dr. Alex Minetto.

Navigation signal analysis and Simulation (NavSAS) research group

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G-PEARL - GNSS Performance analysis, Experimental Assessment and Reporting tool

GNSS Performance analysis, Experimental Assessment and Reporting tool (G-PEARL) is a MATLAB-based standalone application designed to support technical and scientific investigations of GNSS data.

G-PEARL supports GNSS data across multiple data formats. It enables the processing of both raw signal samples and observables, providing automated report generation for a range of scientific analyses, including multi-frequency and multi-constellation GNSS signal acquisition, uncalibrated total electron content estimation, multipath analysis from observables, and state estimation solutions.

This manual documents the functionalities and details of G-PEARL. Please rely on the following to set-up G-PEARL on first use.

License and credits

G-PEARL is released under the **GNU General Public License**, version 3 (GPLv3). See all license texts in '*licenses*' for details. Development and testing activities were performed under the supervision of professor Fabio Dovis. G-PEARL also includes packaged closed software authored by Dr. Andrea Nardin and Dr. Oliviero Vouch.



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Installation Guide – Matlab Runtime Environment

1. Prerequisites

Running **G_PEARL.exe** requires **MATLAB Runtime R2023b (23.2)** installed on your system. The following options can be followed for installation.

The support to automatic report generation requires **Microsoft Word** correctly installed in your system. G-PEARL analysis functionalities are preserved even in case report generation fails.

⚠ Administrator privileges are required to install the MATLAB Runtime.

Option A – Download from MathWorks Website

You can download the Windows version of the **MATLAB Runtime R2023b (23.2)** directly from the official MathWorks website:

<https://www.mathworks.com/products/compiler/mcr/index.html>

For additional details about application deployment and the Runtime environment, refer to the **MATLAB Compiler** documentation under:

Distribute Applications → About Application Deployment → Deployment Product Terms

Option B – Automatic installation from G-PEARL installation wizard

In the G-PEARL file package find *for_redistribution* folder and execute the `MyAppInstaller_web.exe` as administrator.

Follow the wizard instructions to install both the correct MATLAB runtime version and the G-PEARL application in your system.

3. Installation Procedure

1. Install **MATLAB Runtime R2023a through option A or B** (if not already installed).
 2. Find installed `G_PEARL.exe` or place it in your desired working directory.
 3. Run `G_PEARL.exe` as administrator to launch the application.
 4. On first execution, the application may take few minutes to start while the MATLAB Runtime initializes.
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Application startup

1. From the tab “Configuration”, the user can select **RINEX v3.0** observation files containing GNSS observables and **NMEA** files containing information about the receiver PVT solution.
2. After pressing “Load Data”, these files are parsed into internal MATLAB structures and become ready to be processed from the **Analysis** tab. The user can also specify its name or ID at the user textbox which will be included in the reports and choose the output folders where all products will be saved.

IQ samples file have their own dedicated tab which contains both file selection and the experiments. In the current version operation IQS processing is possible only for LuGRE IQS data available at the link.

At the moment of parsing RINEX observation and NMEA log files, the current date and time are saved. When the user generates results, **G-PEARL** creates a folder inside the selected output folder with the date and time saved previously. All results from the current session will be saved inside this folder, organized in sub-folders based on the experiment. For each session, **G-PEARL** keeps for each experiment a counter for how many times the results from that experiment have been saved. The counter is then used to create another layer of sub-folders. For example, results from the first run of experiment 1.a will be saved in “USER_OUTPUT_PATH\CURRENT_DATE\1.a\r01”.

Rinex Observation files

Current version of **G-PEARL** is compatible with **RINEX version 3.04**, which is the latest version supported by MATLAB function `rinexread.m`. The notation of signal bands and channels is also compatible with RINEX 3.04 documentation.

G-PEARL currently supports measurements from GPS, GLONASS, Galileo and BeiDou Global Navigation Satellite Systems.

NMEA

G-PEARL supports as input NMEA log files to plot receiver solutions, the parsing is based on MATLAB function `nmeaParser.m`. All information is parsed from GGA messages, with the exception of the date that is read from RMC messages instead.

Analysis



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All experiments that use GNSS measurements from RINEX observation files are denoted starting with number 1 in sequential order, while those related to PVT solutions from the NMEA log file start with number 2. Experiments can be executed in any order.

Some of the plots generated by G-PEARL have additional features like an interactive legend and custom data tips. In order to use them, please make sure that the folder '`.\library\common`' is included in your MATLAB path.

Interactive Legend

In order to speed up the filtering of only certain time series of interest, the legends can be interacted with allowing for three functionalities:

- Left Click: Toggles visibility of the clicked item
- Shift + Left Click: Inverts the current visibility of all items in the legend
- Double Left Click: Shows only measurements from the bands L1, G1, E1 or B1 (Note: the detection of double click events does not always work accurately)

For example, if the user wants to isolate one time series, they can first click on the corresponding item in the legend to hide it, and then Shift + Left click to invert the selection.

Please note that MATLAB has a limitation of 50 items in the legend, so certain time series may appear in the plot but not in the legend, thus limiting these functionalities.

Custom Datatip

Since measurement plots can include several dozens of time series with similar colors, G-PEARL has an additional feature to help the user identify the correct data. When the user clicks on any point of a time series, a custom data tip appears containing the following information:

- Time – indicates its position on the horizontal axis
- Value - corresponds to the position on the vertical axis
- ID - the name of the item on the legend

To save space in .fig file, these data tips are not a property of the plot, but are instead generated at the moment the user click the plot, so the functionality only works if the associated function is included in the MATLAB path.

Observables

The user can choose to only plot observables from certain constellations using the checkboxes at the top of the tab. These checkboxes can be changed at any time between experiment runs.

1.a Measure signal strength (C/N0)

The experiment outputs one plot with the time series of all measured C/No.

1.b Evaluate signal visibility and PVT availability



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Visibility is intended as the presence of measurements from a given satellite at a given time. Availability is defined as the capability to perform single-point positioning solution at a given time instant, assuming inter-system time offsets are available and do not need to be estimated.

The user can adjust a C/N0 threshold to filter the data. Only satellites that surpass the threshold are counted in the plots described below.

The experiment outputs 3 plots:

1. Time series of the number of unique satellites visible per each band. Unique means that multiple measurement from different channels from the same satellite and the same band are only counted as one (e.g. Galileo E5a and E5b).
2. Time series of the number of unique satellites visible per each constellation. Again, unique means that measurements from different bands for the same satellite are only counted once.
3. Sub-plots of the time series of availability of PVT using only measurements from each constellation, and then with all four as a total. If NMEA log file is loaded and the time axis is matched, the availability of the solution in the file is also evaluated alongside the other subplots.

1.c Measure raw code pseudoranges

The experiment outputs one plot with a time series of all measured code pseudoranges not corrected by the receiver clock.

1.d Evaluate pseudorange errors

The experiment computes and plot the finite-difference third derivative time series of both code and phase pseudoranges. The purpose of this finite-difference is to approximate a discrete-time derivative of third order to remove trends in the pseudoranges due to satellite and receiver movements and obtain a time series with zero-mean. The computation is done while also preserving the same variance of the original time-series.

1.e Measure Doppler shifts

The experiment outputs one plot with a time series of all measured Doppler shifts.

1.f Evaluate the presence and characteristics of multipath

The experiment computes the detrended double-frequency code-minus-carrier (2F CMC) time series, which can highlight the presence of multipath or other sources of error. The two furthest bands for each constellation are considered, namely [L1, L5], [G1, G3], [E1, E5], [B1, B3].

Optionally, the user can decide to turn on the detection of outliers from the time series. If activated, the experiment considers as outliers all values exceeding a certain threshold, chosen by the user as a multiple of the estimated standard deviation on the 2F CMC using a rolling window of size also chosen by the user. The outliers are collected into a table with associated PRN, epoch and the value.



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1.g Evaluate the effect of Ionosphere

The experiment outputs 2 plots:

- Time series of the single-frequency code-minus-carrier (1F CMC) time series, then shifted piece-wise around zero to remove large PRN-dependent offsets due to integer ambiguity in the phase pseudoranges. Changes in the value of 1F CMC are proportional to changes in the Ionosphere delay with a factor of 2.
- Time series of the Slant Total Electron Content (STEC) is estimated for each satellite where signals from at least two frequency bands are available. If possible, the experiment chooses the two bands that are further apart to improve accuracy.

1.h Evaluate the presence of interferences

Receiver state estimation

Information regarding the PVT is parsed from the NMEA log file selected by the user. G-PEARL needs at minimum the GGA messages in order to build internal structures used for plotting. If available, RMC messages are used to obtain a date of the data collection, otherwise a standard date is used.

2.a Characterize Positioning solutions

The experiments outputs two plots:

- A geoplot of the trajectory in Latitude and Longitude coordinates.
- The time series of estimated positions, with three sub-plots for each coordinate. The user can select to use either the Latitude, Longitude and Altitude (LLA) coordinates written in the NMEA log file or the Earth-Centered Earth-Fixed (ECEF) coordinates obtained from converting LLA.

2.b Evaluate Dilution Of Precision (DOP) metrics

The experiments plots the HDOP contained in the GGA message of NMEA log file. Other DOP metrics are included in other NMEA messages that do not contain timestamps, so they cannot be aligned to a common time axis. The figure also includes a second subplot with the number of satellites used.

IQ samples

After choosing the IQ samples file (in .bin extension), the experiment performs probe data and acquisition on the file. The user can select multiple settings for the acquisition:

- Checkbox at the top to choose whether to perform acquisition on the corresponding band.
- Step size of the search in the Doppler domain (50 Hz by default).
- Center frequency of the Doppler search (0 by default).



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- Total Doppler range of the search space (10 kHz by default, which corresponds to ± 5 kHz on each side of the center frequency).
- Number of non-coherent accumulations (1 by default).
- Coherent integration time (length of the code by default).
- False alarm probability (used to set the threshold to declare acquisition).
- Checkbox to choose whether the Doppler shift should be applied to the local code as well.

The settings can be chosen independently for each band. In its current version, G-PEARL performs acquisition on GPS L1 and Galileo E1 bands only.

The experiment plots an histogram of the estimated C/N_0 from the acquisition. Bars corresponding to PRN that are considered acquired are colored. A summary table with the estimated C/N_0 and Doppler shift is also generated.